DGMS 2019 • SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

# Question 75: Electrical Resistivity of Aquifers: Saturation & Salinity Controls (Archie's Law)

Geoelectrical Principles, Pore-Fluid Conduction Mechanisms, and Inverse Physical Relationships

## OFFICIAL EXAMINATION QUESTION

**Consider the following statements regarding an aquifer composed of unconsolidated materials:**

**1.** The resistivity increases with the degree of saturation.

**2.** The resistivity increases with increase in salinity of groundwater.

**Which of the statements given above is/are correct?**

**(a)** 1 only

**(b)** 2 only

**(c)** Both 1 and 2

**(d)** **Neither 1 nor 2 [CORRECT]**

## VERIFIED ANSWER KEY

**Correct Option: (d) Neither 1 nor 2** — Both statements assert the exact opposite of fundamental geoelectrical physics. As demonstrated by Archie's Law:
(1) Electrical resistivity **decreases** as the degree of water saturation increases (because conducting water displaces insulating air).
(2) Electrical resistivity **decreases** as groundwater salinity increases (because higher ion concentration dramatically increases electrolytic conductivity).
Therefore, both Statement 1 and Statement 2 are false.

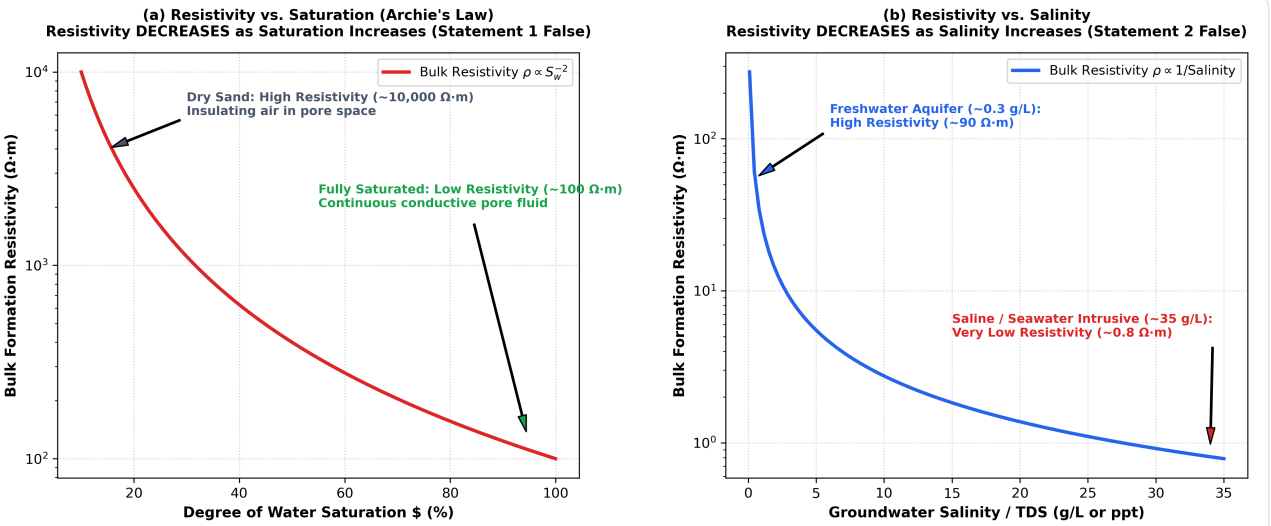



**Figure 75.1 | Archie's Law Controls on Aquifer Bulk Electrical Resistivity:** (a) **Resistivity vs. Water Saturation ($S_w$):** As pore water saturation increases from dry (air-filled) to 100% saturation, bulk resistivity drops by over two orders of magnitude ($\rho \propto S_w^{-2}$). (b) **Resistivity vs. Groundwater Salinity:** Dissolved salts supply charge carriers ($\mathrm{Na^+, Cl^-}$); as salinity increases from freshwater (~0.3 g/L) to seawater intrusion (~35 g/L), formation resistivity plummets from ~100 $\Omega\cdot\text{m}$ down to <1 $\Omega\cdot\text{m}$.

## 1. Geoelectrical Conduction Mechanisms in Aquifers

In unconsolidated sedimentary formations (gravel, sand, silt), common silicate minerals (quartz, feldspar) are electrical insulators with intrinsic mineral resistivities exceeding $10^{12}\ \Omega\cdot\text{m}$.

Consequently, electrical current in an aquifer flows almost entirely by **electrolytic conduction through the interconnected pore fluid**. Bulk electrical resistivity is governed by **Archie's Law (1942)**:

$$\rho = a \cdot \rho_w \cdot \phi^{-m} \cdot S_w^{-n}$$

where:

- $\rho$ is the bulk electrical resistivity of the formation ($\Omega\cdot\text{m}$)
- $\rho_w$ is the electrical resistivity of the pore water ($\Omega\cdot\text{m}$)
- $\phi$ is fractional total porosity
- $S_w$ is the fractional water saturation ($0 \le S_w \le 1$)
- $a, m, n$ are empirical constants ($a \approx 1$, cementation exponent $m \approx 1.3\text{--}2.0$, saturation exponent $n \approx 2$).

## 2. Statement-by-Statement Rigorous Scientific Refutation

### Statement 1: "Resistivity increases with the degree of saturation" — INCORRECT

From Archie's empirical equation:

$$\rho \propto \frac{1}{S_w^n} \approx \frac{1}{S_w^2}$$

- In dry, unsaturated sands above the water table (vadose zone), pore spaces are filled with air, which is an electrical insulator ($\rho_{\text{air}} \rightarrow \infty$). Bulk resistivity is very high ($1,000\text{ to }10,000\ \Omega\cdot\text{m}$).
- As water saturation ($S_w$) increases, continuous films and channels of conducting electrolyte connect the pore network.
- When the aquifer reaches 100% saturation below the water table, electrical resistivity reaches its **minimum value** ($50\text{ to }150\ \Omega\cdot\text{m}$ for fresh sand).
- Therefore, resistivity **decreases** (not increases) as the degree of saturation rises.

### Statement 2: "Resistivity increases with increase in salinity" — INCORRECT

Pore fluid resistivity ($\rho_w$) is inversely proportional to its electrical conductivity ($\sigma_w = 1 / \rho_w$), which is directly proportional to dissolved ion concentration (salinity / TDS):

$$\rho_w \approx \frac{1}{\sum c_i \cdot z_i \cdot u_i} \propto \frac{1}{\text{Salinity}}$$

- In **freshwater aquifers** (low dissolved solids, $\text{TDS} < 500\ \text{mg/L}$), pore water has low conductivity and high resistivity ($\rho_w \approx 20\text{ to }100\ \Omega\cdot\text{m}$). The formation displays high bulk resistivity ($\sim 100\ \Omega\cdot\text{m}$).
- In **saline aquifers** or zones of coastal saltwater intrusion ($\text{TDS} > 10,000\ \text{mg/L}$ to $35,000\ \text{mg/L}$), abundant free ions ($\mathrm{Na^+, Cl^-}$) conduct electricity with minimal resistance ($\rho_w < 0.2\ \Omega\cdot\text{m}$). The formation bulk resistivity drops drastically to **$< 5\ \Omega\cdot\text{m}$**.
- In electrical resistivity tomography (ERT) and vertical electrical sounding (VES), saline aquifers appear as profound **low-resistivity (high-conductivity) anomalies**!
- Therefore, resistivity **decreases** (not increases) as salinity rises.

## 3. Summary Table: Physical Controls on Aquifer Electrical Resistivity

| Aquifer Property | Change in Property | Effect on Bulk Resistivity ($\rho$) | Physical Mechanism |
|---|---|---|---|
| **Water Saturation ($S_w$)** | Increases ($0 \rightarrow 100\ %$) | **DECREASES** ($\rho \propto S_w^{-2}$) | Conductive electrolyte displaces insulating pore air; expands cross-sectional conduction area. |
| **Groundwater Salinity (TDS)** | Increases ($0.1 \rightarrow 35\text{ g/L}$) | **DECREASES** ($\rho \propto 1/\text{TDS}$) | Higher concentration of dissociated ionic charge carriers lowers pore fluid resistivity ($\rho_w$). |
| **Porosity ($\phi$)** | Increases | **DECREASES** ($\rho \propto \phi^{-m}$) | More volume available for conducting pore fluid. |
| **Clay Content** | Increases | **DECREASES** | Surface cation exchange capacity (CEC) provides surface electrical conduction. |
| **Temperature ($T$)** | Increases | **DECREASES** | Viscosity of pore water decreases, increasing ionic mobility by ~2% per °C. |

## 4. Authoritative Textbook References


- **Philip Kearey, Michael Brooks & Ian Hill (2002)** *An Introduction to Geophysical Exploration*, 3rd Ed., Blackwell Science, Chapter 7 ("Electrical Resistivity Methods: Archie's Law and Formation Factors"), pp. 182–195.
- **David Keith Todd & Larry W. Mays (2005)** *Groundwater Hydrology*, 3rd Ed., John Wiley & Sons, Chapter 14 ("Surface Geophysical Investigation: Electrical Resistivity"), pp. 580–595.
- **G. E. Archie (1942)** "The Electrical Resistivity Log as an Aid in Determining Some Reservoir Characteristics", *Transactions of the AIME*, 146(1): 54–62.

- **W. M. Telford, L. P. Geldart & R. E. Sheriff (1990)** *Applied Geophysics*, 2nd Ed., Cambridge University Press, Chapter 8 ("Resistivity Methods").